Road accidents in Italy: where to invest

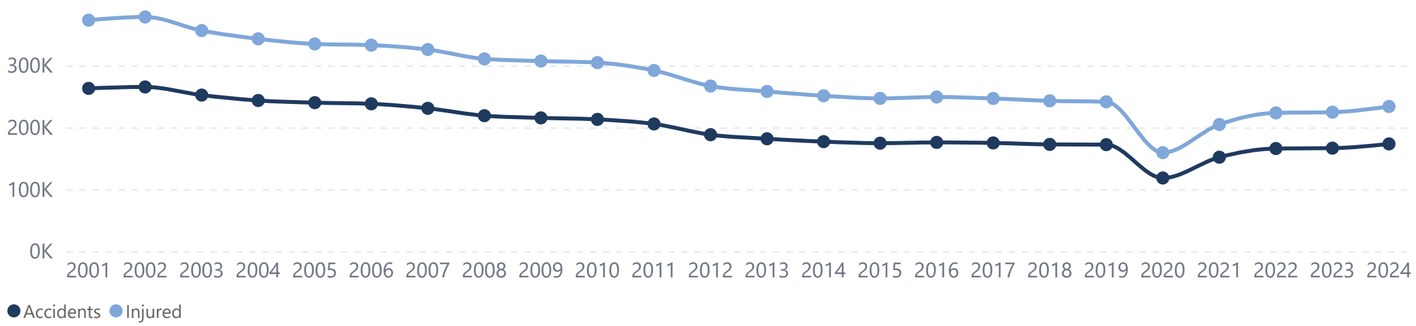
Municipal indicators: yearly average 2022–2023. Source: ISTAT, SITUAS

Region: Tutte

Yearly Accidents	Yearly Deaths	Accidents per 1,000 inhabitants	Deaths per 100 accidents
166.207	3.099	2,82	1,86

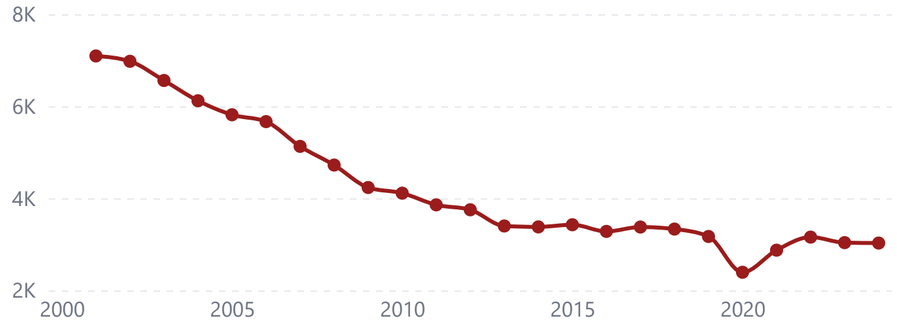
Accidents and Injured people over the Years

National totals, not affected by filters

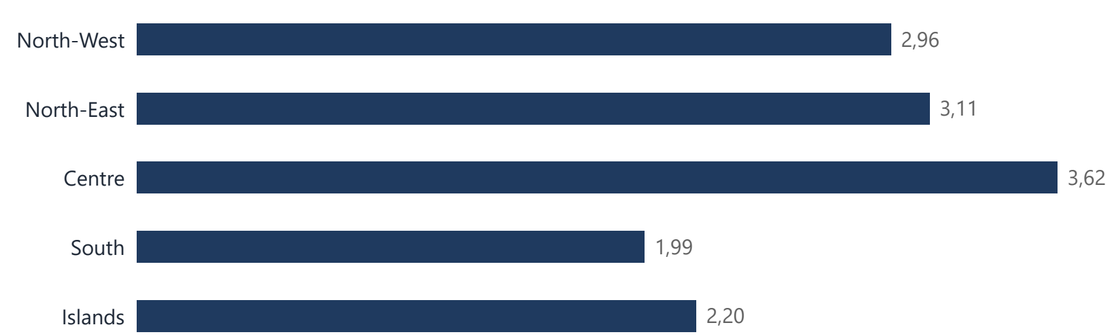


Deaths over the Years

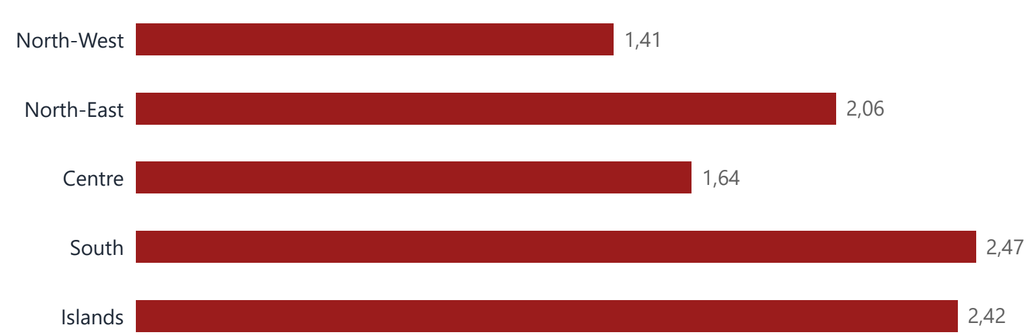
National totals, not affected by filters



Accidents per 1,000 inhabitants by Area



Deaths per 100 accidents by Area



Where to invest

2.180 of 7.900 municipalities shown (≥ 10 accidents per year, 2022-2023)

Municipality: Tutte | Region: Tutte | Cluster: Tutte | Provincial Capital: Tutte

Municipalities by Risk Profile

Colour: cluster. Bubble size: yearly accidents (2022–2023 average)

Dense urban areas · High accident rate · Low frequency, high severity · Low risk, low severity



Top 25 Municipalities by Risk Score

Municipality	Prov.	Cluster	Yearly Accidents	Per 1.000 inh.	Per km²	Risk Score
Bergamo	BG	Dense urban areas	909	7,57	22,59	0,995
Firenze	FI	Dense urban areas	2.604	7,18	25,45	0,994
Pescate	LC	Dense urban areas	25	11,32	11,71	0,993
Forte dei Marmi	LU	Dense urban areas	91	13,43	9,91	0,990
Riccione	RN	Dense urban areas	242	7,02	13,81	0,990
Genova	GE	Dense urban areas	3.790	6,74	15,75	0,989
Assago	MI	Dense urban areas	77	8,35	9,56	0,985
Albissola Marina	SV	Dense urban areas	34	6,60	10,48	0,983
Bolzano/Bozen	BZ	Dense urban areas	659	6,20	12,60	0,982
Milano	MI	Dense urban areas	7.798	5,69	42,88	0,981
Gallarate	VA	Dense urban areas	301	5,67	14,32	0,977
Bologna	BO	Dense urban areas	2.139	5,48	15,19	0,975
Santa Margherita Ligure	GE	Dense urban areas	69	8,12	6,81	0,974
San Benedetto del Tronto	AP	Dense urban areas	271	5,75	10,62	0,974
Lavagna	GE	Dense urban areas	95	7,66	6,81	0,974
Chiavari	GE	Dense urban areas	151	5,49	12,26	0,973
Viareggio	LU	Dense urban areas	344	5,66	10,51	0,972
Grandate	CO	Dense urban areas	20	6,86	6,92	0,971
Livorno	LI	Dense urban areas	912	5,94	8,68	0,971
Curno	BG	Dense urban areas	43	5,71	9,17	0,968
Padova	PD	Dense urban areas	1.057	5,09	11,36	0,965
Recco	GE	Dense urban areas	60	6,37	6,08	0,962
Riva Ligure	IM	Dense urban areas	16	5,58	7,48	0,962
Savona	SV	Dense urban areas	380	6,47	5,81	0,961
Bari	BA	Dense urban areas	1.535	4,85	13,05	0,961

Risk Profiles and Anomalies

Clusters: K-means on 2,180 municipalities. Excess: regression on municipalities with ≥ 5,000 inhabitants

Region: Tutte

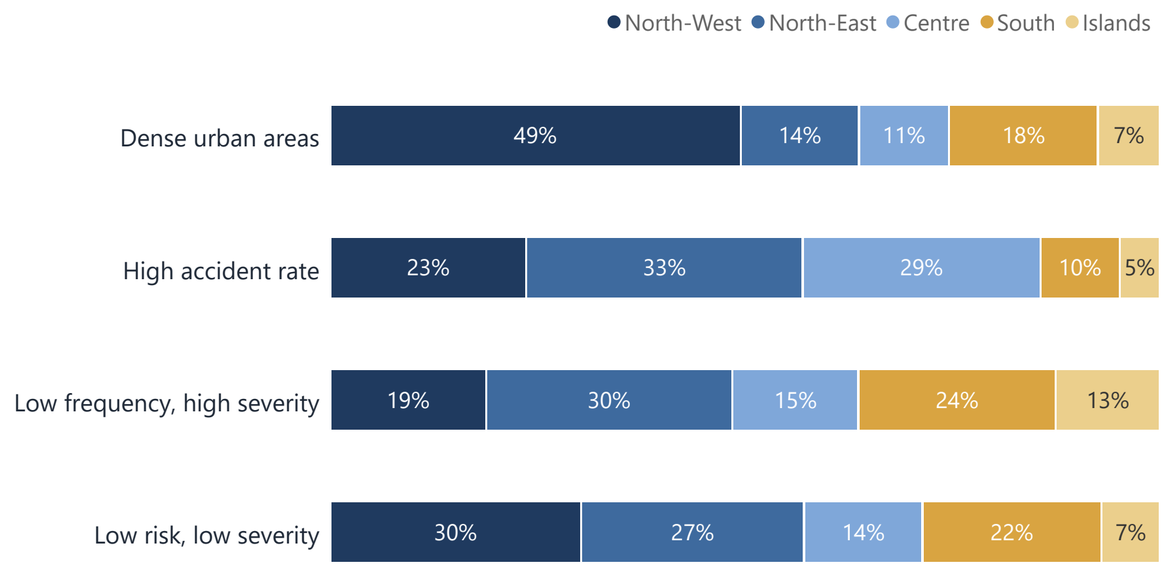
Cluster Profiles

Median Values per Municipality, 2022-2023

Risk Profile	Municipalities	Population	Capitals	Accidents per 1.000 inh.	Accidents per km²	Deaths per 100 acc.
Dense urban areas	294	23955	17,7%	3,29	5,33	0,81
High accident rate	470	11746	7,7%	3,77	1,04	2,21
Low frequency, high severity	793	10364	1,4%	1,99	0,51	4,13
Low risk, low severity	623	8788	1,6%	2,11	0,73	0,00

Where the Clusters are

Share of Municipalities in each Cluster by macro-area



More Accidents than expected

Top 20 by ratio between actual and expected accident rate (regression model)

Municipality	Prov.	Risk Profile	Actual Accidents Rate	Expected Accidents Rate	Actual/Expected
Forte dei Marmi	LU	Dense urban areas	13,43	2,06	6,52
Bentivoglio	BO	High accident rate	11,08	2,07	5,36
Maiori	SA	High accident rate	5,93	1,20	4,95
Porto Cesareo	LE	High accident rate	6,11	1,29	4,72
Assago	MI	Dense urban areas	8,35	1,83	4,57
Santa Margherita Ligure	GE	Dense urban areas	8,12	1,79	4,53
Lignano Sabbiadoro	UD	Dense urban areas	9,20	2,11	4,37
Albissola Marina	SV	Dense urban areas	6,60	1,54	4,30
Vietri sul Mare	SA	Dense urban areas	5,23	1,29	4,06

Conclusions

Risk profiles: group municipalities with similar accident rate, density and severity (K-means, 4 clusters). **Dense urban areas** have the **most accidents**; **low-frequency, high-severity** municipalities have fewer but **deadlier accidents**, likely on extra-urban roads.

Actual / expected: compares each municipality's accident rate with the rate predicted from its population, capital status and area (R^2 = 0.29). A value of 3 means three times the expected accidents.
The **highest values** are mostly **coastal tourist municipalities**, alongside **logistics**, **shopping** and **event hubs**: in any case, far more people and vehicles circulate there than actually live there.

What can be done

- Dense urban areas: **30 km/h zones**, safer crossings and junctions, protected cycle lanes.
- High-severity areas: speed control, roundabouts, safety barriers and **lighting on extra-urban roads**.
- Tourist Municipalities: seasonal enforcement, awareness campaigns for visitors, public transport to the beaches.
- Logistics and commercial hubs: heavy-vehicle routing, **redesigned access junctions**, traffic plans for events.